

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

You have savings of HK\$100,000 and want to invest in two securities listed on the stock exchange of Hong Kong. You have done some research on the internet and have obtained the following information:

State of the economy	Probability of occurrence	Stock A expected return	Stock B expected return
Boom	20%	- 8%	30%
Normal	50%	10%	15%
Recession	30%	15%	- 12%

Required:

- (a) Compute the expected returns on stock A and stock B, respectively. (3 marks)
- (b) Compute the standard deviations of the returns on stock A and stock B, respectively. (3 marks)
- (c) Calculate the composition (i.e., the weights for stock A and stock B, respectively) of the portfolio if you want to achieve an expected return of 9%. (3 marks)
- (d) Calculate the correlation coefficient between stock A and stock B. (4 marks)
- (e) For the "correlation coefficient" between two stocks:
 - (i) Explain what it is and its value range. (4 marks)
 - (ii) Explain its impact on the risk diversification effects on a portfolio comprising two stocks without referring to the above answers calculated in this question. (3 marks)

Question 2 (20 marks – approximately 36 minutes)

Venus Manufacturing Limited ("Venus") is considering two potential orders from customers. Due to limited factory space, Venus can only accept one of these two orders. The information for the two orders is as follows:

	Order A	Order B
Capital investment for specialised machine (Year 0)	HK\$1,200,000	HK\$1,100,000
Incremental cash flows for 4 years		
Year 1	HK\$400,000	HK\$200,000
Year 2	HK\$450,000	HK\$350,000
Year 3	HK\$420,000	HK\$450,000
Year 4	HK\$420,000	HK\$450,000
Scrap value after 4 years (Year 4)	HK\$200,000	HK\$400,000

The capital structure of Venus with relevant market information is as follows:

Common stock: 50 million shares outstanding currently trading at HK\$10 per share.

Preferred stock: 3 million shares selling for HK\$86 per share, with a dividend rate of 6% and a face value of HK\$100.

Debt: Five years ago, Venus issued 300,000 units of 15-year bonds with an annual coupon rate of 4%. The par value of the bonds is HK\$1,000 per unit. The current yield to maturity ("YTM") is 6.5%. The annual interest payment has just been made and the bonds are currently selling for HK\$820 per unit.

Venus is in the 16.5% corporate tax bracket.

Required:

- (a) Venus has just made the annual dividend payment. The dividend payment for the coming year is estimated to be HK\$1, and the growth rate is estimated to be 3% per annum in perpetuity. Calculate the cost of equity for Venus using the dividend discount model. (2 marks)
- (b) Calculate the cost of debt after tax and the cost of preferred stock for Venus. (4 marks)
- (c) Compute the weighted average cost of capital ("WACC") based on the results in (a) and (b). (4 marks)

- (d) Using the result calculated in (c) as the discount rate, compute the NPV for order A and order B respectively and advise Venus on which order should be accepted.

(5 marks)

- (e) Calculate the payback periods for order A and order B, respectively, and determine which order Venus should accept based on this method. Explain TWO disadvantages for using the payback period for project appraisal compared to using the NPV method.

(5 marks)

Question 3 (20 marks – approximately 36 minutes)

Pluto is a manufacturer of Internet of Things ("IOT") products. Any products which are launched after the end of the last financial year are classified as new products. Some details of the activity during the month of October 2023 are as follows:

	Existing Customers	New Customers	Total
Number of Customers	203	15	218
	Existing Products	New Products	Total
Number of units produced	13,000	1,250	14,250
Sales Revenue	HK\$265,500	HK\$37,500	HK\$303,000
Cost of goods sold	HK\$215,055	HK\$27,000	HK\$242,055
Direct labour hours	1,936	201	2,137
Development costs (Note 1)	N/A	N/A	HK\$ 15,075

Note 1: Development costs are for future products and cannot be allocated to the current products of Pluto.

Required:

- (a) Explain the term Key Performance Indicator ("KPI") and identify FOUR elements which are crucial to the development of good KPIs of Pluto. (6 marks)
- (b) A balanced scorecard is a system to measure the performance of a company. Outline the FOUR perspectives from which a balanced scorecard is used to look at companies. (4 marks)
- (c) Recommend and calculate the values for ONE KPI each for any THREE of the four perspectives outlined in (b) that should be addressed by Pluto. (6 marks)
- (d) Explain TWO advantages and TWO disadvantages of Pluto using a balanced scorecard for performance measurement for Pluto. (4 marks)

Question 4 (20 marks – approximately 36 minutes)

Jupiter Limited ("Jupiter") is an original equipment manufacturer ("OEM") selling electronic components and products. In order to upgrade its production capacity, Jupiter purchased a new production line in 2023. The cost of this production line is HK\$200 million, which was 100% financed by a bank loan. The latest financials for the last two years are as follows:

Statements of Profit or Loss for the year ended 31 December

	<u>2023</u> <u>HK\$m</u>	<u>2022</u> <u>HK\$m</u>
Net sales	2,230	2,050
Cost of goods sold	1,710	1,580
Gross profit	520	470
Operating costs	188	145
Depreciation	97	82
Profit before interest and tax	235	243
Interest expenses	56	45
Profit before tax	179	198
Tax expenses	30	33
Net income	149	165
Dividends	45	50
Retained profit	104	115

Statements of Financial Position as at 31 December

	<u>2023</u> <u>HK\$m</u>	<u>2022</u> <u>HK\$m</u>
Non-current assets	2,286	2,176
Current assets		
Inventories	486	375
Accounts receivable	195	160
Cash	141	74
	822	609
Total assets	<u>3,108</u>	<u>2,785</u>
Non-current liabilities	1,000	800
Current liabilities		
Accounts payable	352	339
Other payables	192	186
	544	525
Share capital	1,000	1,000
Retained earnings	564	460
Total equity	1,564	1,460
Total liabilities and equity	<u>3,108</u>	<u>2,785</u>

Required:

- (a) Calculate the following ratios for the years 2022 and 2023, respectively, for assessing the financial performance and position of Jupiter.

Profitability

- i. Gross profit margin
- ii. Return on equity

Liquidity

- iii. Current ratio
- iv. Quick ratio

Gearing

- v. Debt-to-equity
- vi. Interest coverage ratio

(7 marks)

- (b) Analyse the ratios calculated in (a) for the years 2022 and 2023 for the profitability, liquidity, and gearing of Jupiter.

(9 marks)

- (c) Besides return on investment ("ROI"), briefly explain two other common financial measures for performance evaluation of responsibility centres.

(4 marks)

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